DEG2025-5612
Eco-Friendly Biodegradable Bottles and Containers from Banana Flower Stalks: A Sustainable Packaging Solution

Project Title :	Eco-Friendly Biodegradable Bottles and Containers from Banana Flower Stalks: A Sustainable Packaging Solution
Theme :	Waste Management
Project Category :	Hardware model
Abstract :	This project will focus on the development of "biodegradable containers" using "banana stalk fibers, rice husk, and natural gum" as the main materials. The objective is to create an eco-friendly and affordable alternative to plastic packaging, especially for daily-use containers in households. The containers will be used for purposes such as food storage, organizing small items, and general storage needs. The use of banana stalks, an abundant agricultural waste material, and rice husk powder significantly reduces the production cost while still providing a sustainable solution. The natural gum acts as a binder and provides the strength and moisture resistance needed for these containers to be used on a daily basis. The product ensured that the final product is biodegradable and compostable, hence it was a better, safer, and more environmentally responsible alternative compared to traditional plastic containers. Leverage on waste materials to satisfy the evergrowing demand for eco-conscious packaging, this offered a solution that was low-cost, durable, and user-friendly. It encouraged reducing plastic waste through a sustainable circular economy in the packaging industry.
Current Stage of the project :	Prototype Development

Objective of the project :	1. Develop biodegradable containers from banana stalk fibers, rice husk, and natural gum. 2. Offer an affordable, eco-friendly alternative to plastic containers. 3. For durability and resistance to moisture in everyday use. 4. Promote sustainability by creating compostable and non-toxic packaging. 5. Reduce plastic waste and encourage circular economy practices.
Problem statement :	Plastic containers are significant contributors to environmental pollution, since they decompose slowly. Most eco-friendly alternatives available in the market are costlier or fragile. This project seeks to fill this gap by developing low-cost, rugged, biodegradable containers from banana stalks, rice husk, and natural gum. Such containers provide a sustainable alternative to plastic, with the potential for promoting environmental conservation and reducing waste while supporting everyday functionality.
Proposed solution :	The proposed solution will be the utilization of readily available agricultural by-products such as banana stalks and rice husks to develop biodegradable containers. The processing of the materials is done by simple shredding, grinding, and blending witho
Methodology :	1. Material Sourcing: Collect banana stalks, which are very ubiquitous, and rice husks, which are eco-friendly. 2. Pre-Processing: Shred or chop banana stalks by hand and grind rice husks using regular kitchen appliance. 3. Mixing: Mix the materials with a natural binder, such as guar gum, and with water. 4. Molding: Use homemade molds-made plastic, silicone, or cardboard-to shape the mixture into a form. 5. Drying: Let air dry or put in low temperature oven for drying. 6. Wax Coating (Optional): Apply natural beeswax or natural wax to prevent moisture seepage 7. Quality Check: Test durability and

	strength along with moisture resistance of the containers. 8. Packaging: Use environment friendly packaging like recycled paper and cloth.
Working :	This project converts banana stalks, rice husks, and natural gum into biodegradable containers using a simple, low-cost process. The operational process starts with the sourcing of the waste materials: banana stalks are shredded into fibers, rice husks are ground into powder, and natural gum is used as a binder for durability. The materials are mixed into a pulp and poured into molds to form the desired container shapes. After molding, the containers are left to dry and cure, ensuring they are strong and moisture-resistant. Optional wax coatings are applied for enhanced moisture protection, especially for food containers. The process is efficient and scalable, requiring little investment in machinery. Basic molds and drying methods reduce production costs, while using readily available agricultural waste lowers material expenses. This results in an affordable, durable, and eco-friendly alternative to plastic packaging. This method sustains by utilizing waste materials, thereby reducing plastic pollution, and using compostable containers. The project supports a circular economy by utilizing agricultural by-products for repurposing purposes and offering a low-impact solution for packaging needs with minimal environmental footprint and low production effort.
Key features/innovations :	1. Agricultural Waste Utilization: Utilizes banana stalks and rice husks, reducing waste and promoting sustainability. 2. Natural Gum Binder: Uses natural gum for binding, eliminating harmful chemicals and increasing biodegradability. 3. Low Cost Production: Simple techniques such as shredding, grinding, and molding reduce production costs and machinery requirements. 4. Moisture Resistance: Natural gum provides moisture resistance, making containers

	durable for different applications. 5. Biodegradable and Compostable: Containers break down naturally, minimizing environmental impact. 6. Scalable and Local Impact: Easily reproducible in resource-poor settings, creating opportunities in agricultural communities. 7. Modularity: The process of casting enables versatile designs of containers.
Expected outcomes :	1. Economically viable, long-lasting, biodegradable containers. 2. Large reduction in plastic waste and pollution. 3. Circular economy through utilization of agricultural wastes. 4. Development of local economic activities through the product. 5. Durable, moisture-resistant containers, suitable for use in daily lives. 6. High awareness and utilization of environment-friendly alternatives. 7. Helps in packaging sustainability and responsible consumption. 8. Trend toward increased eco-sensitive and more sustainable waste management practices.
Software used :	We haven't used Software.
Hardware used :	Technology Used: 1. Processing of Biodegradable Material: Simple techniques like shredding, grinding, and blending of agricultural waste are used to turn the waste into material for use in containers. 2. Technology of Moulding and Shaping: Different shapes of containers are created by basic molding techniques. This process ensures scalability and cost-effectiveness. 3. Use of Natural Binde: Natural gum is used as a binder. This natural gum provides strength and resistance to moisture without causing harm from chemical byproducts. Hardware Components: 1. Shredder: For tearing banana stalks into fine fibers. 2. Grinder/Mill: For grinding rice husks into a powder for blending with the fibers. 3. Mixing Equipment: For combining banana stalk fibers, rice husk powder, and natural gum into a uniform pulp. 4. Molds: Simple molds

	that can be easily customized according to design requirements for shaping the containers. 5. Drying System: Low-temperature drying ovens or air drying racks for curing the molded containers.
Important design aspect :	1. Raw Materials: Used readily available, biodegradable banana stalks and rice husks. Shred and grind them manually for economy. 2. Binder: Used a natural binder for durability and resistance to moisture. 3. Mixing: Mixed materials uniformly using simple tools such as bowls or hand mixers. 4. Molds: Used DIY silicone or plastic molds to shape the containers. 5. Molding: Compact materials firmly into molds for strength. 6. Drying: slowly under well-ventilated conditions or in a low-temperature oven to avoid cracking. 7. Coating: Used biodegradable wax, if desired, for added moisture resistance. 8. Packaging: Use simple, eco-friendly packaging such as recycled paper. 9. Cost-Efficiency: Use manual tools, having as few expensive pieces of machinery as possible. 10. Eco-Friendliness: Emphasize sustainable materials and processes throughout.
This project gives a :	New solution to a old problem
Write the value addition/advantages over existing solutions :	1.Sustainability: Uses agricultural waste (banana stalks and rice husks), which reduces waste and promotes a circular economy. 2. Cost-Effective: Simple shredding, grinding, and molding reduce production costs, making it more affordable than plastic. 3. Biodegradable: Fully biodegradable and compostable, unlike plastic containers that contribute to pollution. 4. Moisture Resistance: Natural gum provides moisture resistance, ensuring durability for everyday use. 5. Scalable: Low investment, can be initiated in local communities with minimum technological requirements. 6. Health-Safe: Non-toxic and chemical free, making the food safer when stored. 7. Customization: Flexibility and availability of various

	customized molds for all kinds of designs. 8. Economic Growth: Can promote local opportunities as it involves the utilization of waste from agricultural sectors.
Specific areas of application :	1. Food Packaging: Eco-friendly containers for fresh produce, snacks, and takeout. 2. Consumer Goods: Packaging for toiletries, cleaning supplies, and personal care. 3. Retail & E-commerce: Sustainable shipping packaging. 4. Agriculture: Biodegradable seedling trays and plant containers. 5. Hospitality: Disposable plates, cups, and cutlery. 6. Healthcare: Storage containers for medical supplies. 7. Office & Stationery: Organizers and storage boxes. 8. Pharmaceuticals: Biodegradable pill bottles. 9. Cosmetic Packaging: Eco-friendly containers for cosmetics and skincare.
Impact of the solution :	1. Environmental Impact: Reduces plastic waste, prevents pollution, and promotes a circular economy by using agricultural waste. 2. Social Benefits: Offers non-toxic containers with safer food storage and increases eco-awareness. 3. Economic Benefits: Low-cost production generates local employment and promotes small enterprises. 4. Benefits to Industry: Provides a renewable alternative to plastic, supporting industries in fulfilling eco-friendly requirements and regular.
Commercial viability :	1. Target Markets: Environment-sensitive consumers, packaging of food and beverages, supermarkets, hospitals, hotels, cosmetic industries. 2. Unique Selling Proposition : Biodegradable, low cost, agricultural-waste-based potted containers produced from banana plant stalks or rice husk, moisture resistance, non-toxic. 3. Revenue Model: Products to be sold to businesses or B2B contracts for responsible packaging. 4. Cost: Low costs for materials as well as processing. 5. Selling Price: Compares favorably with plastic ones. 6. Investment:

	Initial setting up for production and raw material, low-cost machinery.
Do you seek incubation support from DIPEX? :	Yes